Harish Alakuntla

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SUMMARY

AI/ML professional with a strong foundation in **Machine Learning, Deep Learning, NLP, Generative AI, and RAG**. Skilled in **Python, Scikit-learn, TensorFlow, SQL, LangChain, and LLMs**, with hands-on experience building and deploying practical AI solutions.

EDUCATION

Data Science with GEN AI, *Innomatics Research Labs* Hyderabad, Telangana
Coursework: Python, Data Analysis, EDA, SQL & Power BI, Machine Learning, Deep Learning, Generative AI & Agentic AI

B.Tech-CIVIL ENG, *Netaji Institute of Engineering and Technology* Hyderabad, Telangana
Percentage - 67%

SKILLS

Programming & Libraries:
Python, OOP, NumPy, Pandas, Matplotlib, Seaborn

Machine Learning
Supervised & Unsupervised Learning, Regression, Classification, Decision Trees, Random Forest, XGBoost, KNN, SVM, Ensemble Learning, Cross-Validation, Hyperparameter Tuning, SMOTE

NLP
Text Preprocessing, Tokenization, Stemming, Lemmatization, POS Tagging, NER, BoW, TF-IDF, N-grams, Word2Vec, GloVe, FastText, RNN, LSTM, GRU, Attention, Transformers, BERT

SQL + Power BI
SQL, Joins, CTEs, Subqueries, Window Functions, Power BI, DAX, Data Modeling

Deep Learning
ANN, CNN, Transfer Learning, Activation Functions, Backpropagation, Batch Normalization, Dropout, Regularization

GenAI
LLMs, Prompt Engineering, Embeddings, LangChain, RAG, Vector Databases, Agentic AI

PROJECTS

Vehicle Fuel Economy Prediction, *Highway MPG Prediction using Machine Learning*
- Built an end-to-end Machine Learning regression model to predict highway fuel economy (MPG) using vehicle specifications such as engine displacement, cylinders, transmission, drivetrain, fuel type, and model year.
- Performed data preprocessing, One-Hot Encoding, Power Transformation, feature scaling, model comparison, and RandomizedSearchCV hyperparameter tuning.
- Developed a Random Forest Regressor achieving 0.969 R^2 and 1.10 MPG RMSE, and deployed the model using Streamlit.

Diabetes Prediction Using ANN, *Deep Learning–Based Diabetes Risk Classification*
- Built an ANN classification model to predict diabetes using 100K+ patient records with demographic and clinical features.
- Performed EDA, data preprocessing, categorical encoding, scaling, and class-imbalance handling to prepare the data.
- Applied regularization, dropout, batch normalization, and hyperparameter tuning to improve model generalization.
- Evaluated the model using Accuracy, Precision, Recall, F1-score, and ROC-AUC, achieving ~96% test accuracy and 0.95 ROC-AUC.

Tech Stack: Python, Pandas, NumPy, Scikit-learn, TensorFlow/Keras, Matplotlib, Seaborn, Optuna.

Financial Services AI Assistant Chatbot, *Domain-Specific Conversational AI using LangChain & OpenAI*
- Built a domain-specific conversational AI chatbot using LangChain, OpenAI, and Streamlit to handle financial-services queries related to loans, credit cards, fees, eligibility, and payments.
- Implemented prompt templates and conversational memory to maintain context across user interactions and generate relevant responses.
- Designed a domain-restricted chatbot workflow that provides financial information while avoiding unrelated queries, and deployed it through an interactive Streamlit interface.

Languages

- English
- Telugu
- Hindi